SAIRAM SRIDHARAN

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EDUCATION

Northeastern University | Boston, MA **May 2026**
Masters of Robotics, Electrical and Computer Engineering, **GPA: 3.919**
Related Coursework: Robot Sensing and Navigation, Mobile Robotics, Robot Mechanics and Control, Field Robotics, Assistive Robotics, Reinforcement Learning, Optimization Algorithms, Non-Linear Controls.

Vellore Institute of Technology | Chennai **May 2024**
Bachelor of Technology, Mechatronics and Automation, **GPA: 9.12/10**
Related Coursework: Industrial Automation, PLC, Machine Learning, Computer Vision, IoT, Product Design and Development.

TECHNICAL SKILLS

Programming: Python, C++, Rust, MATLAB, SIMULINK
Software and Tools: ROS1/ROS2, Linux, Gazebo, Isaac Sim, MuJoCo, OpenCV, PyTorch, Docker, Git, SolidWorks, Raspberry Pi
Robotics: SLAM, Visual Odometry, Sensor Fusion, Path Planning, Motion Planning, Computer Vision, Reinforcement Learning

WORK EXPERIENCE

Research Intern | WIKA | Chennai **June 2022 - July 2022**
- Assisted the team in design and production of Industrial sensors, including flow, temperature, and pressure measurement devices.
- Contributed to calibrating Industrial sensors to maintain the accuracy and precision of pressure and temperature measurement instruments.

Research Intern | Lucas TVS | Chennai **May 2022 - June 2022**
- Monitored and supervised assembly lines for alternators, starters, motors, and wipers used in automotive applications, gaining practical experience in production workflows and process optimization.
- Worked with the R&D department to research product innovations and evaluate emerging technologies for future development.

PROJECTS

Autonomous Jenga-Playing Robot | Northeastern University | Robotic Manipulation – Jenga
- Built a robotic system using the WidowX-250 arm to autonomously play Jenga by integrating perception, control, and mechanical design for real-world manipulation tasks.
- Implemented AprilTag-based block detection, real-time pose transformation, and closed-loop feedback control in ROS 2, achieving successful block extraction without tower collapse.

Camera–LiDAR Extrinsic Calibration On-the-fly | Northeastern University | Online Camera–LiDAR Calibration
- Developed an online camera–LiDAR extrinsic calibration pipeline using edge-based alignment of projected 3D LiDAR features with 2D image edges.
- Validated robustness through synthetic offset injection and recovery experiments, demonstrating suitability for real-world robotic deployment.

Structure-from-Motion (SfM) 3D Reconstruction Pipeline | Northeastern University | SfM 3D Reconstruction
- Built a complete SfM pipeline from scratch, including feature extraction, epipolar geometry, pose recovery, triangulation, PnP and global bundle adjustment using GTSAM.
- Validated reconstruction accuracy by comparing outputs with COLMAP and extracting real camera intrinsics (K) and radial distortion (k_1) from COLMAP calibration.

Adaptive Path-Guided RRT (APG-RRT) Implementation | Northeastern University | APG-RRT Path Planning
- Implemented and evaluated APG-RRT as an efficient alternative to RRT, generating smoother, shorter, and collision-free trajectories in structured environments (corridors, bends, narrow passages).
- Extended the algorithm from 2D Python simulations into a ROS 2 framework, integrating with TurtleBot3 in Gazebo to validate autonomous navigation in realistic robotic scenarios.

Three-Link Planar Biped Walking using Hybrid Zero Dynamics | Northeastern University | 3-Link Biped Control
- Modeled a three-link planar biped and derived hybrid walking dynamics for underactuated locomotion.
- Designed stable walking gaits using virtual constraints, Bézier polynomials, zero dynamics, and feedback linearization

Route Master: Navigating Dynamic Environments with Advanced Mapping | VIT University | Autonomous Mobile Robot
- Developed a cost-effective autonomous mobile robot using LiDAR and Raspberry Pi, integrated with ROS for both tele-operated and autonomous navigation.
- Performed iterative testing to achieve reliable performance in dynamic indoor environments, with the robot effectively avoiding obstacles and following predefined paths with high precision.